

Symbolic Geometric Context Keeper

Technical Architecture & Mathematical Specification

An Alternative to Dense Embedding Compression (RAG) for Exact Intent Representation

1. System Vision

Traditional Retrieval-Augmented Generation (RAG) fails to capture true intent because it compresses structured semantic relationships into flat, fuzzy, low-dimensional coordinate neighborhoods. This specification outlines an alternative **Neuro-Symbolic Vector Space Architecture**. The system preserves absolute symbolic identities without compression, using sparse infinite-dimensional linear algebra and Outer Tensor Products to systematically bind entities, roles, and temporal events into a clean mathematical truth layer.

2. Mathematical Foundations

2.1 The One-Hot Symbolic Basis (No Compression)

To avoid the semantic bleeding inherent to dense embeddings, every distinct entity symbol and semantic dimension name discovered in a conversation is assigned its own independent, mutually orthogonal (perpendicular) base coordinate axis. This forms a dynamically scaling **Sparse One-Hot Basis Vector Space**.

Given an expanding vocabulary universe, individual tokens map cleanly as:

Entity Space (E): $\mathbf{e}_{\text{Alice}} = [1, 0, 0, \dots]^T$, $\mathbf{e}_{\text{PostgreSQL}} = [0, 1, 0, \dots]^T$

Role Dimension Space (D): $\mathbf{d}_{\text{proposer}} = [0, 0, 1, \dots]^T$, $\mathbf{d}_{\text{subject}} = [0, 0, 0, 1, \dots]^T$

2.2 Context Binding via Outer Tensor Products

When an entity fills a specific dimension role within an event, the binding is established mathematically through an **Outer Product Matrix**. This matrix represents an uncorrupted structural 'peg connection' in the memory graph:

$$\mathbf{M}_{\text{Fact}} = \mathbf{e}_{\text{Entity}} \otimes \mathbf{d}_{\text{Role}}^T$$

For example, if *Alice* is bound to the role of *proposer*, the resulting structural coordinate matrix activates an exact, noise-free scalar junction at the exact intersection of row index *Alice* and column index *proposer*, yielding a sparse incident block containing zero compressed ambiguity.

3. Event Representation: The Switchboard Network

An event cannot be characterized as a single fluid curve or flat point. Geometrically, an event is an **incident constellation network** constructed by compiling the exact summation of all active structural fact matrices occurring at a given conversation timestamp (t_n).

The Event Tensor Equation:

$$\mathbf{E}_n = \sum (\mathbf{e}_{\text{Entity}_i} \otimes \mathbf{d}_{\text{Role}_i}^T)$$

3.1 Geometric Collapsing & Diverging

- Collapsing:** Higher-level macro operations project the dense, high-dimensional multi-entity event matrix down into a unified topological step junction along a discrete event tracking timeline (Z-axis). This encapsulates complete textual sentences into compact structural keys.
- Diverging (Exploding):** When localized inspection is triggered, the engine resolves the specific timeline node back out along its explicit basis directions, immediately rendering the pure, uncompressed micro-details of the underlying

conversational state.

4. Graph Topology and Query Framework

4.1 Mathematical Coordinate Collisions

Since entities share an absolute index coordinate, overlapping topics across different events act as deterministic bridges. If *Event 0* binds *PostgreSQL* as a proposal subject, and *Event 1* binds *PostgreSQL* as a test baseline, the *PostgreSQL* basis index is activated across both independent event tensors. Relationships are extracted via direct index routing paths rather than approximate proximity checks.

4.2 Deterministic Representation of Ambiguity

Ambiguity is not represented by a fuzzy, averaged vector. Instead, it is modeled as an exact **Set Superposition Matrix**. If an LLM states 'They approved it' and cannot determine whether 'They' refers to *Alice* or *Bob*, both discrete coordinates are flagged with partial weight allocations simultaneously:

$$\mathbf{v}_{\text{They}} = 0.5 \cdot \mathbf{e}_{\text{Alice}} + 0.5 \cdot \mathbf{e}_{\text{Bob}}$$

The query engine tracks this spatial uncertainty by identifying variance spreads across coordinate matrices, triggering contextual ambiguity-generated prompts to settle structural forks when necessary.

5. Engine Input Data Interface

The upstream LLM acts as an uncompressed structural parser, passing discrete JSON schemas to the geometric coordinate mapper. The following payload demonstrates the exact mapping boundary for *Event 0*:

```
{
  "event_id": "event0",
  "temporal_position": "t0",
  "mappings": [
    {"entity": "Alice", "dimension": "role_in_event", "value": "proposer"},
    {"entity": "PostgreSQL", "dimension": "proposal_subject", "value": "proposed_option"},
    {"entity": "payments platform", "dimension": "proposal_target", "value": "target_system"}
  ]
}
```